Caps-TripleGAN: GAN-Assisted CapsNet for Hyperspectral Image Classification

Xue Wang, Kun Tan, Qian Du, Yu Chen, Peijun Du

Abstract— The increase in the spectral and spatial information of hyperspectral imagery poses challenges in classification due to the fact that spectral bands are highly correlated, training samples may be limited, and high resolution may increase intraclass difference and interclass similarity. In this paper, in order to better handle these problems, a Caps-TripleGAN framework is proposed by exploring the 1-D structure triple generative adversarial network (TripleGAN) for sample generation and integrating CapsNet for hyperspectral image classification. Moreover, spatial information is utilized to verify the learning capacity and discriminative ability of the Caps-TripleGAN framework. The experimental results obtained with three real hyperspectral data sets confirm that the proposed method outperforms most of the state-of-the-art methods.

For the published version of record document, go to:
http://dx.doi.org/10.1109/TGRS.2019.2912468